

Akshat Gupta

Software Engineer — Backend, Full-Stack & Observability

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PROFESSIONAL SUMMARY

Software Engineer and final-year B.Tech (CSE-AI) student specializing in backend microservices, observability, and full-stack development. Currently a Software Developer at Mark AI, building ML-driven screen recommendations and an AI booking agent for a DOOH advertising platform; previously at VibeMonitor, building AI-driven test automation, high-throughput log-ingestion pipelines into ClickHouse, and a unified monitoring platform across 5+ microservices with Prometheus, Grafana, Datadog, and New Relic. Comfortable owning systems end to end — from FastAPI/Python and React/Next.js to AWS, Kubernetes, and CI/CD — with proven delivery as a full-time engineer, freelancer, and edtech startup co-founder.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, Kotlin, Swift, C#

Backend & APIs: FastAPI, Node.js, Express.js, Prisma, REST APIs, OAuth 2.0, JWT, WebSocket, LangGraph, RAG, OpenAI API, Amazon Bedrock, Whisper, ONNX Runtime

Frontend: React.js, Next.js, Angular, Redux, Tailwind CSS, Material-UI, Vite, Electron, AOS, EmailJS

Databases & Storage: PostgreSQL, MongoDB, ClickHouse, Redis, MongoDB Atlas, Firebase / Firestore

Observability & Monitoring: New Relic APM, Datadog, Grafana, Prometheus, OpenTelemetry, AWS CloudWatch, Loki, Tempo, Mimir, SigNoz, RED & USE Methodologies

Cloud & DevOps: AWS (IAM, RDS, CloudWatch, EKS, Lightsail, SSM Parameter Store), Google Cloud Platform, Docker, Docker Compose, Kubernetes, Caddy, CI/CD Pipelines, Load Balancing, hPanel

Developer Tools & Integrations: Git, GitHub Apps, Postman, Swagger, Prisma Studio, pgAdmin4, incident.io, Notion, Android Studio, Playwright, Stripe, Razorpay, Twilio

EXPERIENCE

Mark AI | Software Engineer | Full-Time | Platform

Jun 2026 – Present

Advertisement-booking platform for digital out-of-home (DOOH) displays.

- Leveraged **ML-based recommendation models** to suggest **related screens** and **competitor screens** to advertisers, improving campaign targeting and screen discovery across the inventory.
- Built an **AI agent** that converts **natural-language** requests into executable booking scripts, letting users book screens easily through conversational input instead of manual workflows.
- Developed a responsive **React** frontend with an interactive **Three.js** experience, integrating **Blender**-created 3D assets for immersive screen visualization.
- Built a native **Android** signage-player **APK** in **Kotlin** that self-registers to the in-house **CMS** portal and streams scheduled content to **DOOH** displays over a persistent **WebSocket** connection to the **FastAPI** backend for real-time playback and remote control.
- Owned the **CMS** portal and its connected main site end to end — building both so operators and advertisers manage screen inventory, content, and campaign scheduling from a single platform.
- Deployed the platform on **Google Cloud Platform (GCP)**, configuring a **load balancer** to distribute traffic across instances for reliable, scalable production serving.

ESG Ratings | Full Stack & AI Engineer | Freelance | Platform

Jul 2026 – Present

AI-driven ESG rating platform — sustainability scoring for corporates and BFSI lenders.

- Revamped the entire platform end to end and **migrated the whole backend stack to Python** — replacing the legacy **PHP** calculators with a single **FastAPI** service on **MongoDB**, and rebuilding the public site, both calculators, and the superadmin dashboard in **Next.js 16** (App Router), **React 19**, **TypeScript**, and **Tailwind CSS v4**.
- Built **multiple AI agents**, one for every scenario — a page classifier, a **KPI evidence-scoring engine**, report narrative and rating-driver writers, and separate BFSI category, keyword, and qualitative agents — fanned out concurrently across report pages with a thread pool and **SHA-256** caching so repeat analyses cost nothing.
- Defined the **KPI framework** and authored and corrected the **scoring formula**: every KPI scored **0–100** on the evidence found across all three pillars, best-evidence-wins aggregation, **evidence-type** and **sector-materiality** weighting, **35/30/35 E/S/G** pillar weights, then transition and controversy adjustments and rating safeguards down to a final grade.
- Built both public calculators — the **ESG calculator** and the **BFSI calculator**, which adds loan-type weightings and a lending recommendation — ingesting **PDF** and **DOCX** sustainability reports and returning a graded, per-KPI breakdown.
- Designed and built the whole **UI** — public site, blog with a **TipTap** editor, and an admin dashboard with **Chart.js** score visualizations, drag-and-drop ordering, and design tokens for brand color, type, and motion.
- Built the **editable rating report** — a client-branded **.docx** ESG Rating Summary generated from the stored scores with AI-written narrative, strengths, weaknesses, and rationale, plus in-browser **PDF** export.
- Deployed on **AWS** — **Lightsail**, **MongoDB Atlas** over **IAM** authentication, and secrets in **SSM Parameter Store** — with **Docker Compose** and **Caddy** auto-TLS behind a one-command deploy script, and made the scoring model switchable between **OpenAI** and **Amazon Bedrock** from the dashboard.

Observability & monitoring platform — work spans backend, frontend, observability, and infrastructure.

- Built **Beryl**, an AI-driven end-to-end test-automation product that crawls a target application from a single URL — including pages behind authentication — generates test cases in natural language, converts them into **Playwright** scripts, and runs them in a headless (browserless) browser to report pass/fail results and an overall site-reliability score.
- Instrumented services with custom **Prometheus** metrics and alerting rules following the **RED** (Rate, Errors, Duration) and **USE** (Utilization, Saturation, Errors) methodologies, giving the team a consistent, end-to-end view of service health.
- Built a high-throughput **log ingestion platform** with custom **SDKs in Java, Python, and JavaScript** that batch and asynchronously stream high-volume logs into **ClickHouse**; tuned ingestion and storage around ClickHouse's columnar storage engine, compression, and binary format to keep it fast and space-efficient.
- Built a centralized **observability platform** that unifies **AWS CloudWatch, New Relic APM, Datadog, and Grafana** across **15+ microservices**, correlating logs, metrics, and traces and adding **role-based access control (RBAC)** and custom pipelines for centralized monitoring and alerting.
- Developed an automated **Root Cause Analysis (RCA)** engine on **LangGraph** that uses agentic workflows to trace incidents through repository commit history and code changes, cutting incident investigation time by **65%**.
- Owned Beryl's post-signup execution layer, scaling **Playwright** runs across a worker pool and building its frontend; focused on **self-healing selectors** and eliminating flaky tests so a test that passes once keeps passing on every run.
- Worked across the stack — designing **RESTful APIs in FastAPI and Python** with a clean, layered architecture, and building the **Next.js** frontend with responsive, optimized rendering and efficient client-side state management for real-time monitoring views.
- Ran container orchestration on **Kubernetes (Amazon EKS)**, handling service scaling and deployment workflows across multiple environments.
- Integrated **AR visualization** using **MindAR** and **3D human models**, enabling clients to preview advertisements in real-world environments through marker-based scanning before campaign deployment.
- Improved **CI/CD** pipelines with multi-stage workflows, build caching, and artifact reuse, reducing build times by **20%** across **10+ pipelines**, and automated **RDS** snapshot management for reliable database backups and environment migrations.

Sakhi Women | Full Stack Developer | Freelance | Platform

Jan 2026 – Feb 2026

Women-only ride-booking startup — subscription rides, dynamic pricing, and real-time WhatsApp coordination.

- Built a women-only ride-booking platform end to end with **Next.js, Express.js, and MongoDB**, including role-based **user** and **admin** dashboards and a responsive, reworked UI.
- Implemented the full ride lifecycle on the admin dashboard — accepting or declining incoming ride requests and assigning each booking to drivers, with live status updates when a driver accepted a ride.
- Built **WhatsApp**-based **OTP** login and a templated notification flow on the **Twilio WhatsApp API** that keeps riders informed at every step — booking confirmation, driver assignment, and driver-accepted alerts.
- Designed **MongoDB** indexes to keep queries fast as usage grew, cutting query latency by roughly **70%** under **1,500–2,000** concurrent users.
- Built the pricing engine with distance-based (per-km) fares for one-way, return, and round trips, **subscription** plans, and night-time **dynamic pricing** (1.5x); launched a **coupon** system and integrated a **payment gateway** for ride payments.

Mark AI | Full Stack Developer | Freelance | Platform

Dec 2025 – Jan 2026

Advertisement-booking platform for digital out-of-home (DOOH) displays.

- Built an advertisement-booking platform for digital displays end to end — **Next.js** frontend, **FastAPI** backend, and **MongoDB** — with role-based **admin** and **user** dashboards, a reworked UI, and a tested end-to-end booking flow.
- Integrated the **Screenox** CMS portal and **REST API** into the app for real-time screen-inventory management and campaign scheduling across **100+ digital displays**.
- Integrated **Razorpay** for payments and built a **coupon** / discount system for ad bookings.
- Configured a **load balancer** with **round-robin** distribution across multiple **FastAPI** instances using **connection pooling** and **async** request handling, and added **MongoDB** indexing to keep booking workflows fast under concurrent load.

iTax Easy | Full Stack & Server Developer | Internship | Platform

Feb 2025 – Sep 2025

Tax-filing platform — GST and ITR compliance across web and mobile.

- Built high-performance web and mobile applications for the tax-filing platform using **PostgreSQL, Next.js, Prisma, and Android Studio**, integrating the **Sandbox API** for **GST** and **ITR** filing; cut load times by **45%** and increased user retention by **30%**.
- Redesigned the **database architecture**, accelerating query response times by **60%** and reducing server costs by **30%**.
- Designed and managed role-based dashboards for **super-admin, admin, and user** roles, owning access control and operations across all three.
- Owned hosting, deployment, and server management on **hPanel**, including configuring a **load balancer** on a dedicated port to handle production traffic.
- Built an **eLibrary** module that aggregates publicly available records from government portals and presents them within the platform.

PrepSaarthi | Co-Founder & Technical Lead | Platform

Nov 2024 – Jul 2025

Mentor-driven edtech startup — career guidance and student upskilling.

- Co-founded and led engineering at **Prepsaarthi.com**, a mentor-driven edtech startup for career guidance and student upskilling, owning the full technical roadmap and generating **INR 1 Lakh** in revenue.
- As **technical lead**, built and owned the entire frontend and backend — shipping **5+ core features** with **React.js** and **MongoDB**, then migrating the frontend to **Vite** for faster builds and load times, which lifted user engagement by **40%**.
- Built a **counsellor booking system** with student–mentor scheduling and a **ratings and reviews** feature.
- Built the payment flow with an automatic **commission deduction** on each booking, splitting the platform’s cut from the counsellor’s payout.
- Implemented **SMS-based OTP** authentication, role-based **admin and user dashboards**, and secure **RESTful APIs**, and deployed the platform on **Hostinger (hPanel)**.

Al-Zira Technology Private Limited | Frontend Engineer & Mentor | Internship

Feb 2024 – Dec 2024

Service-based software company — frontend development and UI engineering.

- Delivered multiple client-facing frontend projects in **React** as part of a service-based team, building responsive, interactive user interfaces.
- Engineered complex, smooth UI **animations** and motion effects to create polished, engaging experiences across projects.
- Promoted from **Engineer** to **Mentor** after roughly 5–6 months, guiding and reviewing the work of other frontend engineers.

PROJECTS

Loupe — Screen Recorder & Video Editor | Electron, Swift, .NET 8, WebCodecs, Whisper, ONNX[GitHub](#) | [Live](#) | Aug 2026 – Present

- Building a cross-platform desktop screen recorder for **macOS** and **Windows** that **zooms while recording** — hold a modifier and scroll to zoom up to **4x** with the view tracking the cursor — shipped as a **DMG**, a **Homebrew** cask, and a Windows installer.
- Wrote the native capture layer twice behind one protocol — **Swift (ScreenCaptureKit, CGEventTap, AVFoundation)** on macOS and a single **.NET 8** program (**Windows.Graphics.Capture**, low-level input hooks, **Media Foundation**) on Windows — so the **Electron** main process drives four identical helpers on both platforms.
- Built the non-linear **editor**: a timeline with cut, trim, and undo history, speed ramps from **0.25x–8x** using a **WSOLA** time-stretch that holds voices at natural pitch, and annotations, webcam bubble, keystroke overlay, wallpaper backgrounds, and transitions composited as layers.
- Added **on-device captions** — **Whisper** through **transformers.js** and **ONNX Runtime** in a worker, with word-level timings, automatic language detection, and **SRT/VTT** export — plus an audio chain with **RNNoise** denoising, music ducking, and voiceover recording. Everything runs locally: no account, no upload.
- Built the **export pipeline** — **MP4**, **WebM**, and **GIF** muxing at **1080p/1440p/4K** and **60 fps**, with bitrate planning that hits either a quality target or a hard file-size limit — and covered the app with **60+ unit test files** and **Electron** end-to-end suites.

Cosmic-Trek (Orrery App) | Three.js, WebGL, D3.js, React, Redux, RAG

[GitHub](#) | [Live](#) | Dec 2025 – Mar 2026

- Built an interactive **3D solar-system simulator** with **Three.js** and **WebGL**, featuring **8+ celestial bodies** and **50+ exoplanets**, and cut rendering time by **60%**.
- Programmed a **physics-based satellite-launch simulator** with **5+ orbital patterns**, reducing computation load by **30%**.
- Created realistic **collision and gravitational simulations** with **10+ custom impact effects**, increasing user engagement by **75%**.
- Built a **Retrieval-Augmented Generation (RAG)** chatbot in **React**, grounded strictly in the *Orbital Mechanics* (Stanford) textbook — it answers only from that source and declines questions outside the book’s scope.

UpInTheSky | React, Firebase, Firestore, JavaScript, Tailwind CSS

[Live](#) | Aug 2025 – Oct 2025

- Built the complete **React** frontend and UI for a travel-booking platform, designing reusable components and a **responsive** layout with **Tailwind CSS** across every page.
- Implemented the booking experience — letting users browse and explore curated travel packages, view destination details, and complete bookings through a clean, end-to-end flow.
- Integrated **Firebase** as the backend — **Firestore** for real-time data and **Firebase Authentication** for secure login — serving **100+ travel listings** across multiple destinations.

Legal E-Library | Angular, Node.js, Express.js, Prisma ORM, PostgreSQL, CSS

[GitHub](#) | Mar 2025 – Jun 2025

- Built a digital library for legal case laws with **advanced search and filtering**, improving research efficiency by **40%**.
- Designed a responsive UI in **Angular**, giving seamless access to **10,000+ legal documents** across **50+ categories**.
- Built the backend **RESTful API** with **Node.js**, **Express.js**, and **Prisma ORM** over **PostgreSQL**, optimizing query performance by **60%**.

SipCraft | HTML, CSS, JavaScript, Stripe

[GitHub](#) | Oct 2024 – Dec 2025

- Built a complete **e-commerce** website from scratch in **vanilla HTML, CSS, and JavaScript** with no frameworks or libraries, including a responsive product catalog and storefront UI.
- Implemented the core shopping flow in plain JavaScript — an **add-to-cart** button, a dynamic cart with item quantities and live total updates, and a full **checkout** flow — driven entirely by direct DOM manipulation.
- Integrated **Stripe** to handle secure online payments end to end, taking an order from cart through to a confirmed purchase.

EDUCATION

KIET Group of Institutions (AKTU)

B.Tech in Computer Science and Engineering (Artificial Intelligence) | GPA: 7.8/10

Ghaziabad, Uttar Pradesh
Nov 2022 – May 2026 (Expected)

- **Relevant Coursework:** *list your actual courses*

Mahabharati International School (CBSE)

Senior Secondary (Class XII) — 85%

Kota, Rajasthan
2020 – 2022

- Cleared **JEE Main** with an All India Rank of approximately **60,000**.

Secondary (Class X) — 87%

2019 – 2020

LEADERSHIP & ACHIEVEMENTS

- **Co-Founder & Technical Lead**, Prepsaarathi.com — a mentor-based platform for career guidance and student upskilling.
- **Tech Lead**, DevOps Club, KIET Group of Institutions — guiding members on DevOps, CI/CD, and cloud practices.
- Member of **FOSSCU**, an open-source software community, engaging with open-source projects and initiatives.